PRODUCT

Pixel Lab

Evolving Game Assets with Genetic Algorithms + AI

IAT 460 - Computational Arts Final Project

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The Problem

Manual Design is Slow

Game devs need hundreds of unique sprites, drawing each takes weeks

Random Generation is Ugly

Pure randomness produces noise, not usable art

AI Image Gen Lacks Control

Neural networks generate but don't let you steer the direction

Solution

1

Genetic Algorithm

Evolves assets through crossover, mutation, and tournament selection

2

Claude AI (LLM)

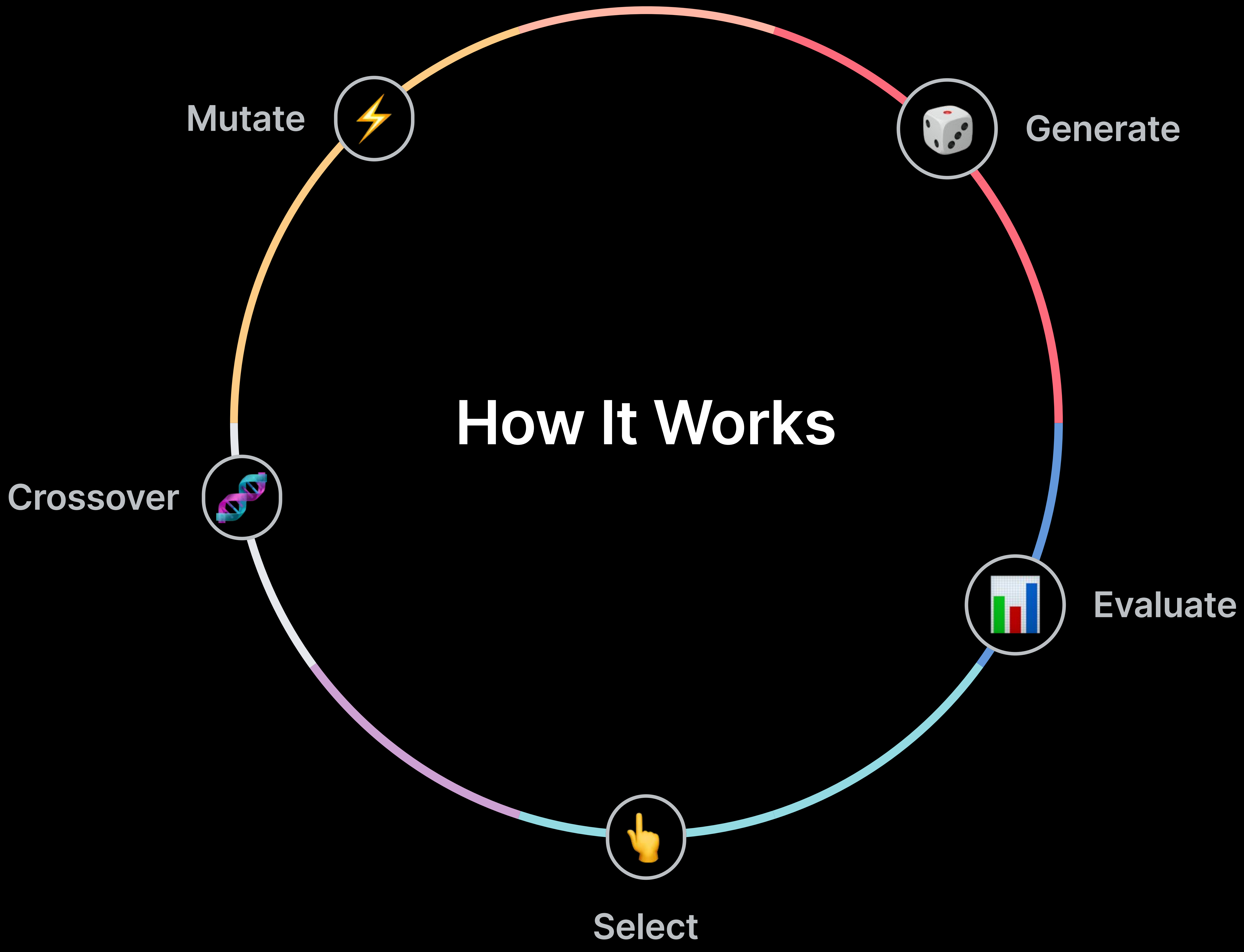
Generates unique names, personalities, and RPG stats from traits

3

9 Fitness Agents

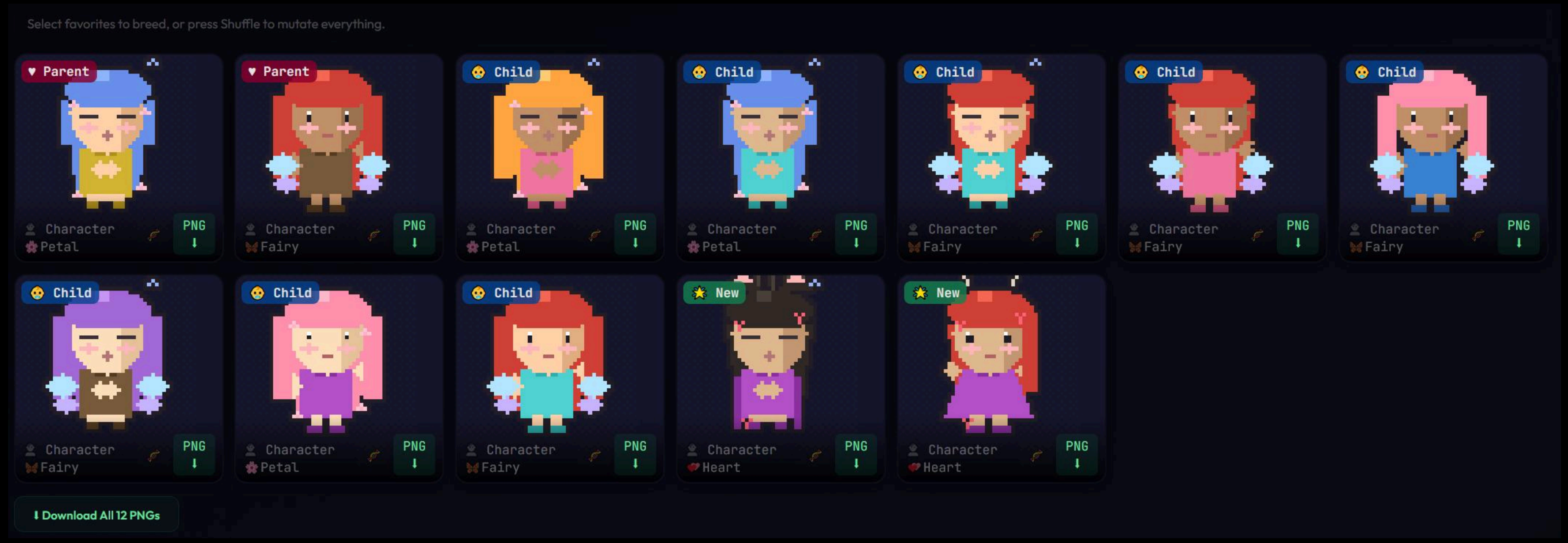
Opposing pairs with adjustable weight sliders shape evolution

Genetic Algorithm



Select 2 parents - breeding will mix their traits

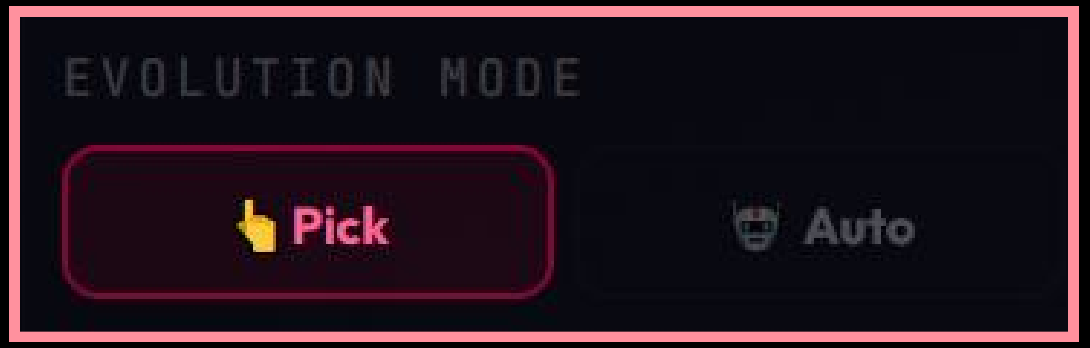
Children inherit hair, outfit, pose, and effects from both parents (Parent/Child/New badges)

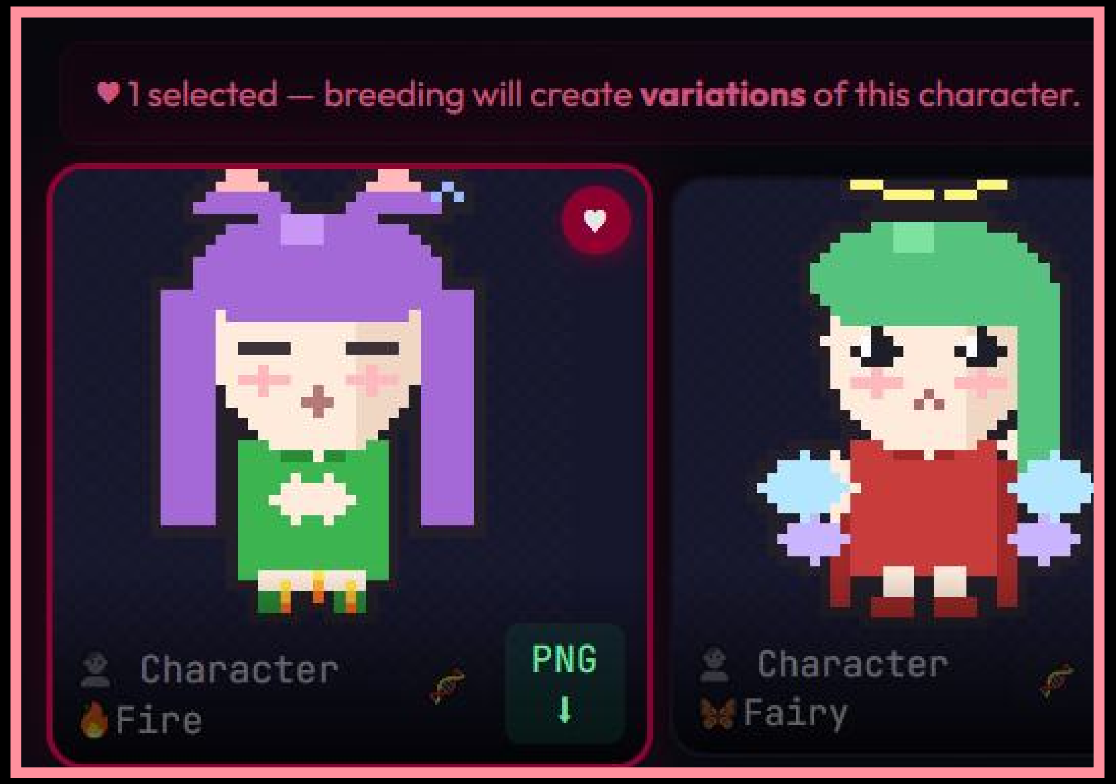


Three AI Techniques Combined

Genetic Algorithm

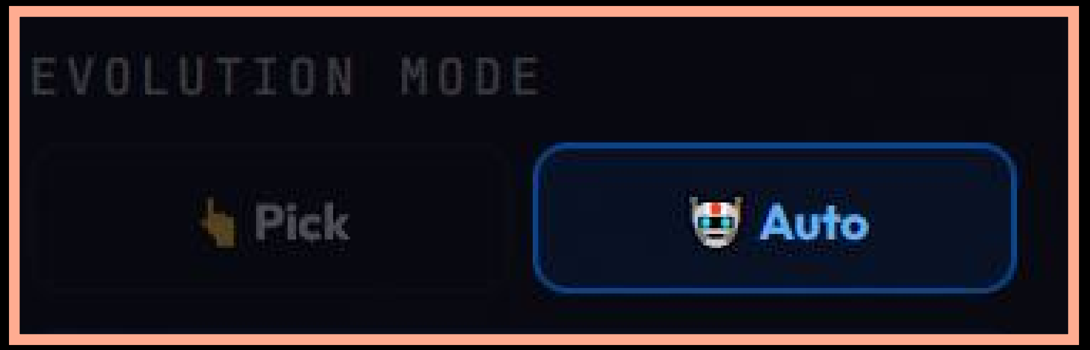
Crossover + Mutation evolve 12 parameters per character. Tournament selection picks fittest parents.

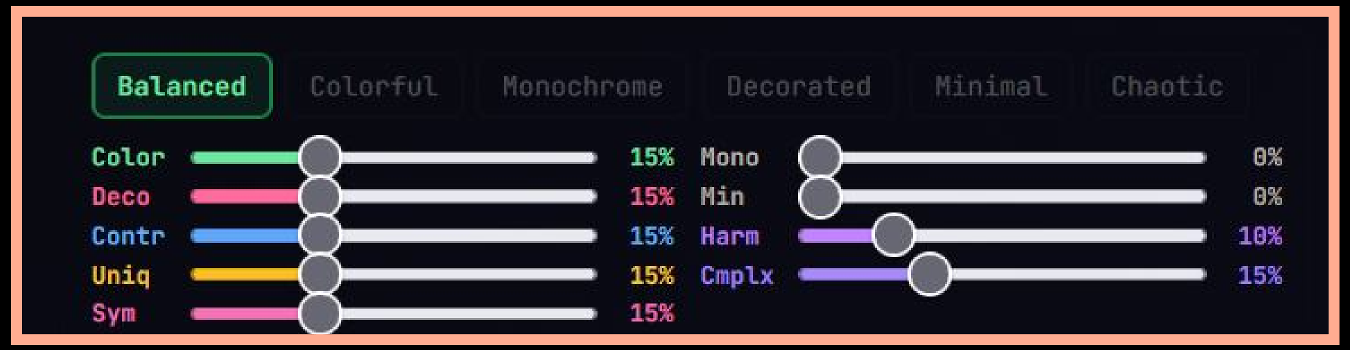




9 Fitness Agents

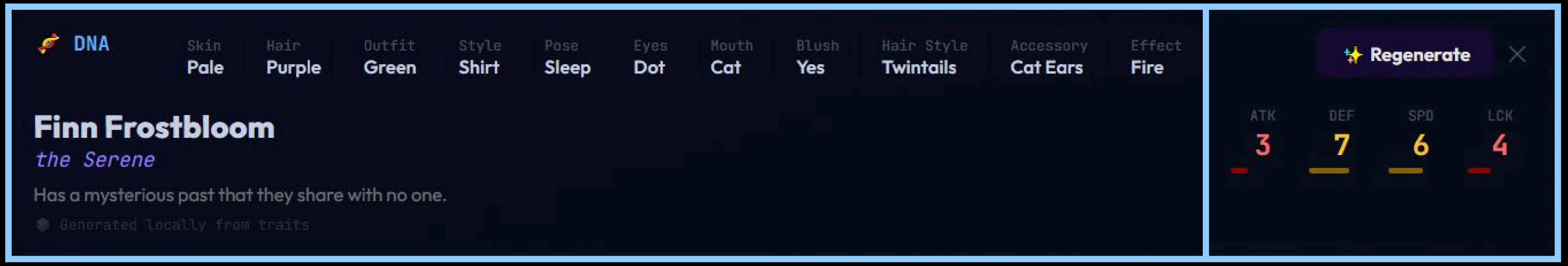
Opposing pairs (Colorful vs Mono, Decorated vs Minimal) with adjustable weight sliders. Fitness sharing prevents convergence.





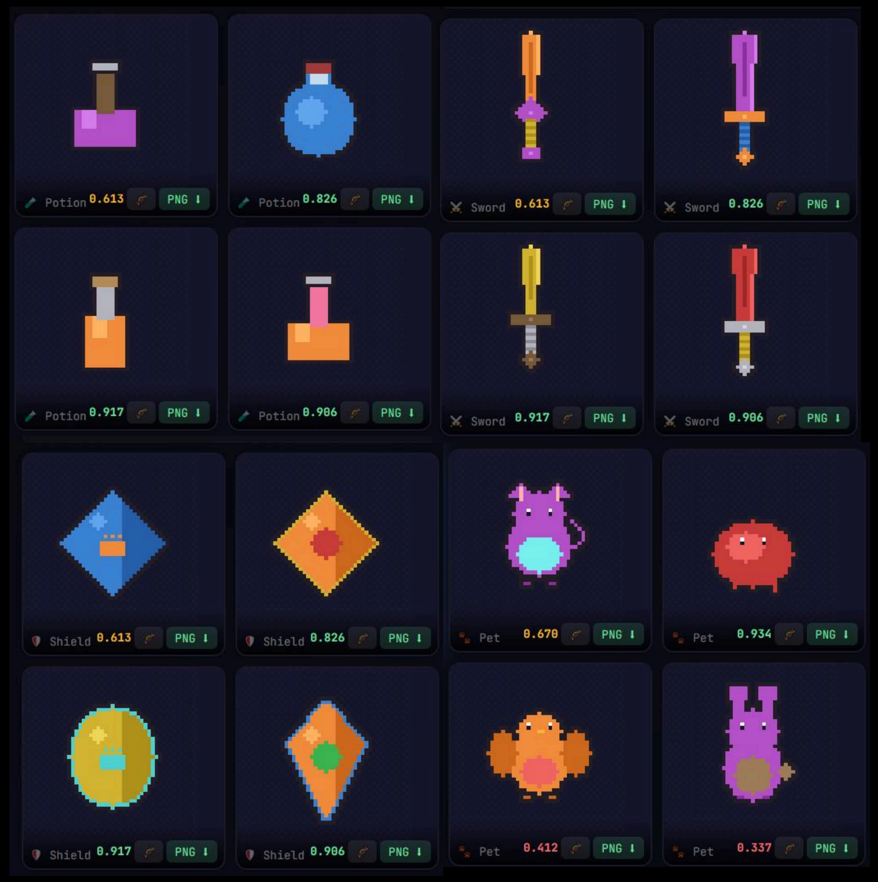
Claude AI (LLM)

Reads character traits and generates unique name, title, personality, and RPG stats (ATK/DEF/SPD/LCK). Two AI systems collaborate.

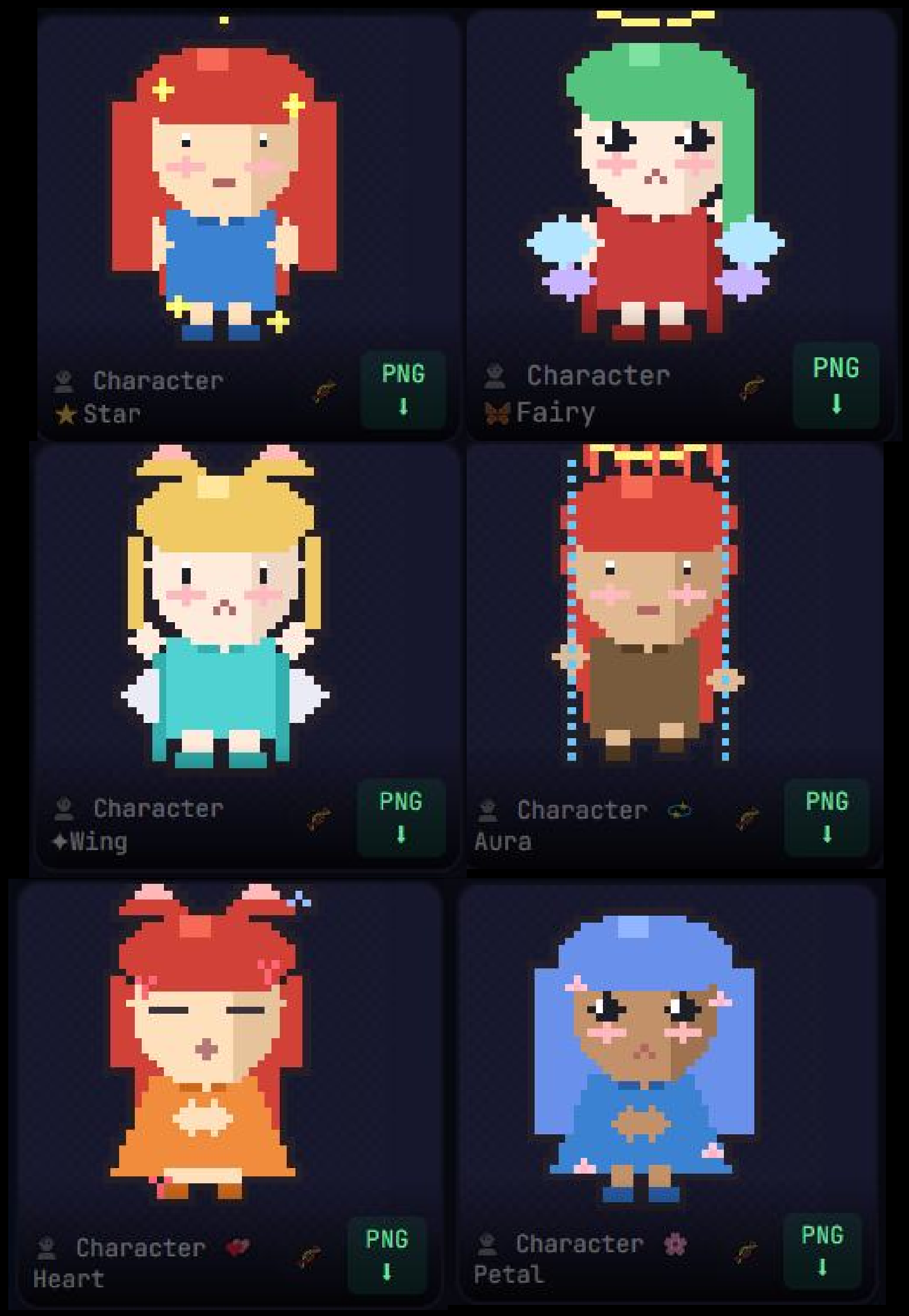


System Features

6 Categories



12 Effects



Palette System

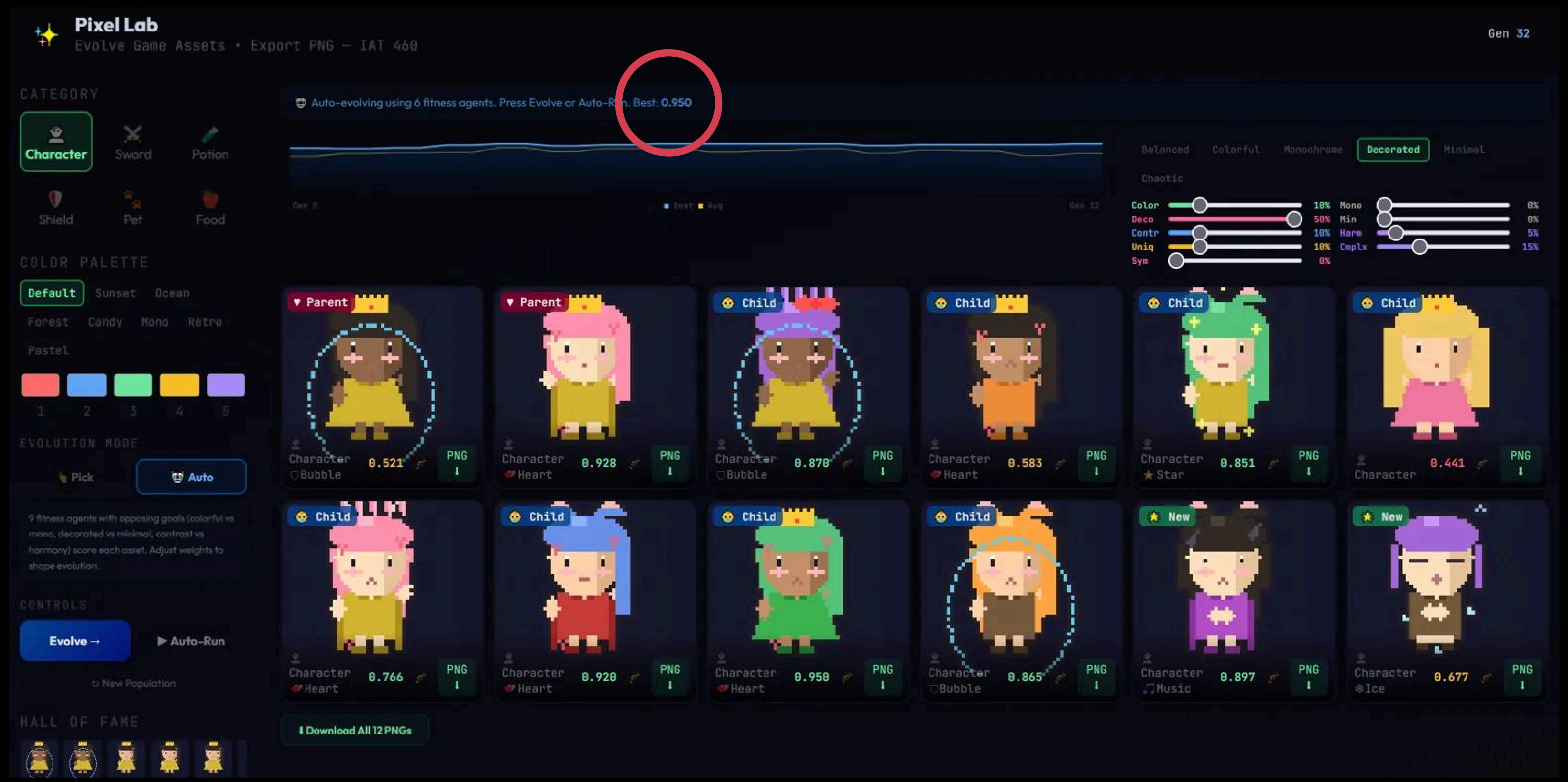
PNG Export

Live Demo

https://youtu.be/TH-owmpUB2U?si=vBlhOVklFYfurAKY

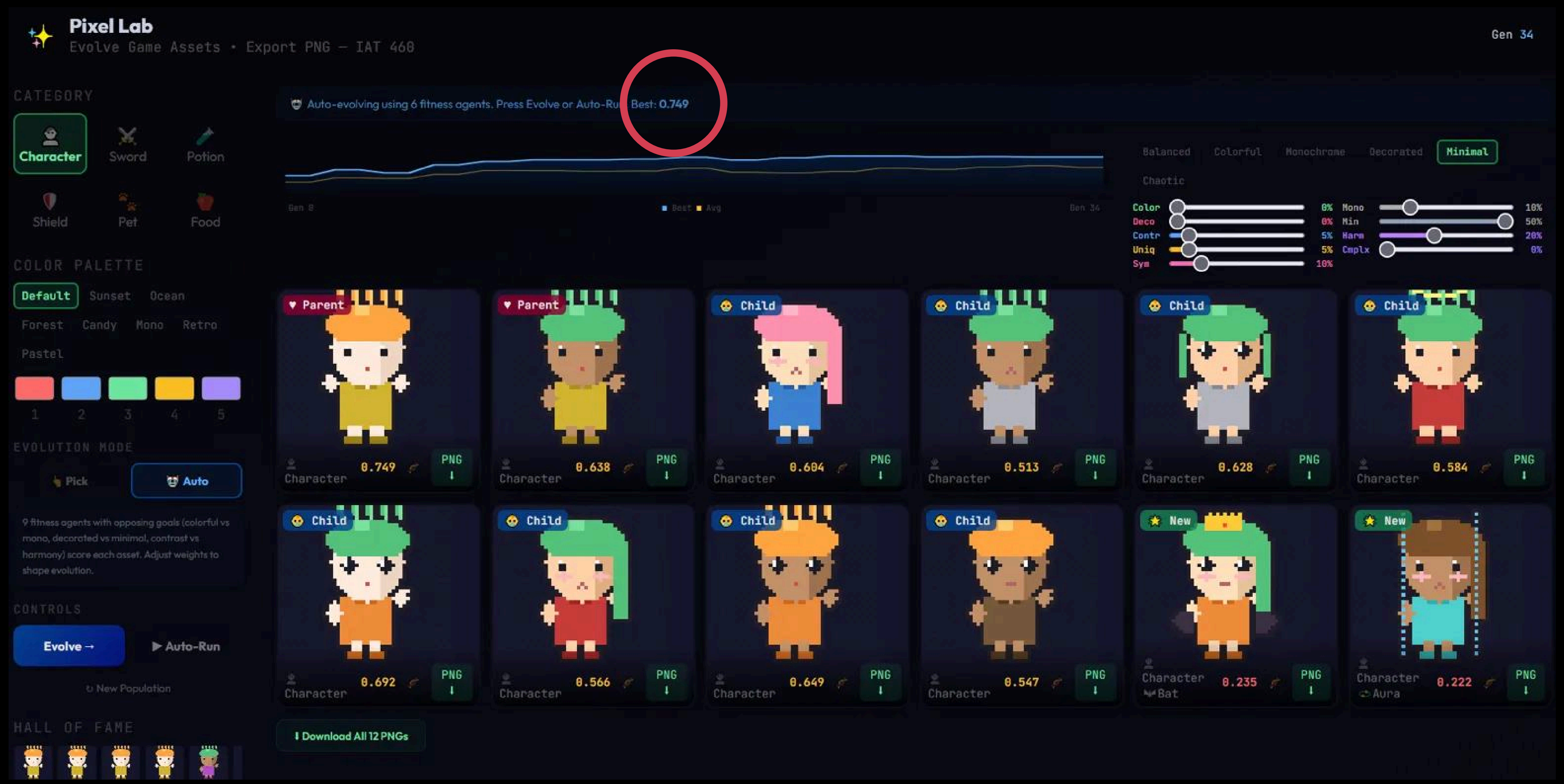
Results - Fitness Weights Shape Output

Decorated Preset (50% Deco weight)



Crowns, hearts, wings, effects
Best fitness: 0.95

Minimal Preset (50% Min weight)



Clean characters, no accessories
Best fitness: 0.75

Challenges & Solutions

Convergence	Problem: All characters looked the same → Fitness sharing penalizes similar genomes
Pixel Mutation	Problem: Raw pixel evolution can't find structure → Template genome - parameters, not pixels
Similar Agents	Problem: Changing weights didn't change output → Opposing agent pairs (Colorful vs Mono, Decorated vs Minimal)
No Visible AI	Problem: Users couldn't see what the algorithm does → DNA viewer, fitness scores, weight sliders, AI naming

Conclusion & Future Work

Key Findings

- Template-based genomes + GA produce recognizable, exportable game assets in minutes
- Opposing fitness agents (Decorated vs Minimal) produce visibly different aesthetic outcomes under the same algorithm
- Human selection → more aesthetically pleasing results; Auto selection → more diverse population
- Combining GA (visuals) + LLM (narrative) creates a richer creative system than either alone
- Fitness definition is the single biggest factor in shaping creative output

What's Next

- Higher resolution sprites (64×64, 128×128) with sub-pixel shading
- Sprite sheet animation export (idle, walk, attack, hurt)
- Neural style transfer to apply artist styles to evolved output
- Multi-user collaborative evolution (vote-based breeding like Picbreeder)
- Game engine integration (Unity/Godot drag-and-drop plugin)

Thank You